On the Future of Education in the Age of AI

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Abstract

In this paper, we study how elite institutions select people once generative AI can do much of the technical work that has historically been the main route of entry for outsiders. Technical skill can be checked by anyone, whatever the candidate's background. The skill that matters after AI is judgment, which is harder to observe, and information about it comes from referees who are not disinterested. We first show that letters from referees at other institutions carry no information in any equilibrium, because those referees want their students placed regardless of quality, while sponsors inside the institution share in its reputation and transmit information credibly, but with a bias. The information an institution holds about outsiders is therefore a garbling of its information about insiders. We then show that, as AI makes checkable skills less valuable, outsiders are under-selected even by unbiased evaluators, any given favoritism matters more, the apprenticeships that once trained and revealed talented outsiders disappear, and winning a patron gets easier. AI can push the other way if it is used to verify ideas: a simple pass–fail test to which the outsider's home institution commits in advance places outsiders more often than full disclosure would. Finally, because reputation is shared among the members of an institution, these forces erode quality slowly and invite entry by institutions that evaluate outsiders through verifiable channels.

JEL: D82, D83, I23, J24, J71, O33. **Keywords:** artificial intelligence, Blackwell informativeness, cheap talk, Bayesian persuasion, influence activities, apprenticeship, collective reputation.

1 Introduction

Elite intellectual institutions, such as leading research departments, top law firms, strategy consultancies and research laboratories, perform two functions. They produce ideas, and they certify those who produce them. The second function rests on a technology of selection, and the openness of such institutions to outsiders depends on what that technology can observe and, more subtly, on whose reports it can believe.

Historically, the main route of entry for outsiders has been *verifiable* human capital. Technical competence, such as the ability to run an empirical design, solve a model, or implement an estimator correctly, has three properties that make it an unusually democratic signal. It can be demonstrated within a short relationship, its assessment consumes little of the evaluator's attention, and it leaves little room for discretion, since its quality can be checked. The apprenticeship structure of elite training can be understood as an institution organized around these properties. For instance, many economics PhD students now begin as pre-doctoral research assistants, whose job is largely to clean data and write code for faculty members. Such positions turn a candidate's technical skill into a credible signal whatever her origin, and once she is inside, they expose her to the tacit knowledge and judgment of the institution.

Generative AI is, to a first approximation, a technology for automating such execution. As its capability rises, the return to execution skill falls, and institutions select increasingly on *conceptual capital*: the ability to identify questions worth asking, to frame them productively, and to recognize which of many feasible projects matter. Conceptual capital differs from execution skill in exactly the dimensions that governed mobility. It is assessed subjectively and through sustained interaction, and the people who hold the relevant information, such as advisers, letter writers and colleagues, are not disinterested.

In this paper, we develop a theory of how this shift changes selection into elite institutions. Our starting point is that the information an institution holds about a candidate is not a primitive but an equilibrium object. Information about conceptual capital is soft: it is held privately by referees and transmitted through cheap talk, and whether it is transmitted at all depends on the referee's incentives. A referee at an outside institution cares about placing her student and has no stake in the elite institution's reputation. Her preferences over the institution's decision do not depend on the candidate's quality, so her soft information does not affect the outcome in any equilibrium. A sponsor inside the institution shares in its reputation. Her preferences depend on the candidate's quality, and she transmits information credibly, although with a bias. The insiders' informational

advantage is therefore not merely a matter of proximity. It is a consequence of incentives, and it takes the precise form of a Blackwell ranking: the institution's information about outsiders is a garbling of its information about insiders.

We show that this ranking has sharp implications once the verifiable component of a candidate's value recedes. Four mechanisms emerge.

1. **Informational exclusion.** A Blackwell-superior experiment generates a distribution of posterior means that is a mean-preserving spread of the inferior one. At selective thresholds, the better-observed group is over-represented, even when ability distributions coincide and evaluators are unbiased. Verifiable execution skill, which is observed equally for all, dilutes this asymmetry, and AI removes the dilution (Theorem 1, Proposition 2).

2. **The favoritism multiplier.** Sponsors' bias operates on the soft component. Measured relative to the remaining dispersion of the selection index, a given premium matters more after AI. In our numerical example, full automation doubles the standardized size of any given premium (Proposition 3).

3. **The collapse of apprenticeship.** Apprenticeship once *formed* outsiders' conceptual capital, and it also *revealed* it, by giving outsiders a sponsor. AI dissolves the demand for apprentices. The resulting group difference in conceptual capital is genuine, endogenous and self-confirming in the sense of Coate and Loury (1993) (Propositions 4 and 5).

4. **The cheapening of patronage.** Candidates close to sponsors can win favor through influence activities (Milgrom, 1988). Because AI compresses the dispersion of the selection index, the effort needed to bridge a given gap in rank falls. Before AI, an elite institution admits conceptual talent, execution talent and a few exceptionally skilled cultivators of patrons; after AI, it admits conceptual talent and a broader class of moderately skilled cultivators (Propositions 6 and 7).

These forces are not inevitable. AI is also a potential *verifier* of ideas, and verification is precisely what outsiders lack. When an AI-based evaluation is publicly observed, it adds a common experiment for both groups, and the insiders' advantage in terms of appointments vanishes as its precision grows (Proposition 8). More strikingly, if an outsider's home institution can design a verifiable test of ideas and commit to it, the logic of Bayesian persuasion (Kamenica and Gentzkow, 2011) implies that the optimal test is a coarse pass–fail standard that places outsiders strictly more often than full disclosure would, whereas any amount of soft information places them, effectively, never (Proposition 9).

The central variable is thus verifiability, not precision. Whether AI opens or closes elite institutions depends on whether it is used as a substitute for execution or as an instrument of verification.

Finally, we embed selection in a model of collective reputation (Tirole, 1996). Each member bears only a fraction of the reputational cost of a weak appointment, so the equilibrium premium for one's own candidates exceeds the institutional optimum and increases with the size of the institution. Reputation responds with a lag, which protects incumbents, and the formation channel partly masks the decline: insiders really are better on average, even as the institution selects less well than it could. Entry by institutions that commit to verifiable evaluation is the eventual corrective.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 studies communication and derives the Blackwell ranking. Section 5 establishes informational exclusion and the favoritism multiplier. Sections 6 and 7 develop the apprenticeship and influence channels. Section 8 studies AI as a verifier, including the persuasion problem. Section 9 analyzes reputation. Section 10 derives empirical implications, and Section 11 concludes. Proofs are in the Appendix.

2 Literature Review

Our paper builds on the literature on statistical discrimination. Phelps (1972) and Aigner and Cain (1977) show that employers who observe some groups' productivity less precisely treat these groups differently even without prejudice. Cornell and Welch (1996) show that screening with group-specific information leads to under-selection of the less well-observed group at selective thresholds. Coate and Loury (1993) show that group differences can be self-confirming when they discourage investment. Our paper differs from these papers in that the precision ranking is not assumed but derived from communication incentives, and we cast it in terms of Blackwell experiments (Blackwell, 1953). Compared to these papers, one of our main innovations is to show how a technological shock to the composition of valued skills activates, amplifies and entrenches the ranking.

Our communication analysis draws on the cheap-talk model of Crawford and Sobel (1982) and on the comparison of delegation and communication within organizations by Dessein (2002). Our analysis of verifiable tests draws on Bayesian persuasion (Kamenica and Gentzkow, 2011), on its applications to grading and certification (Ostrovsky and Schwarz, 2010; Boleslavsky and Cotton, 2015), on competition among senders (Gentzkow

and Kamenica, 2017), and on the comparison of distributions through rotations (Johnson and Myatt, 2006). Our treatment of the manipulation of AI evaluation relates to Frankel and Kartik (2019). The analysis of influence activities follows Milgrom (1988) and Milgrom and Roberts (1988), signal jamming follows Holmström (1999), and signaling follows Spence (1973). Our model of reputation follows the collective-reputation model of Tirole (1996).

On the economics of AI, we adopt the task-based perspective of Acemoglu and Restrepo (2018) and the distinction between prediction and judgment of Agrawal et al. (2019). Korinek (2023) discusses the use of AI in economic research specifically. On gatekeeping in science, our theory is consistent with evidence on favoritism and information in editorial decisions (Laband and Piette, 1994; Brogaard et al., 2014), on social ties in economics (Colussi, 2018), on expertise and bias in peer review (Li, 2017), on the concentration of publication and promotion (Heckman and Moktan, 2020), and on the entrenchment of scientific elites (Azoulay et al., 2019).

3 The Model

3.1 Candidates and Technology

An elite institution fills a mass K of positions from a population of candidates. Each candidate belongs to a group $g \in \{I, O\}$. *Insiders*, of mass m_I, are affiliated with a member of the institution, and *outsiders*, of mass m_O, are not. A candidate is characterized by a state $\omega = (\theta, a)$, where θ denotes conceptual capital and a denotes execution skill. To isolate the informational mechanisms, we initially take ability distributions to be identical across groups; Section 6 relaxes this. Unless stated otherwise,

$$\theta \sim N(0, \tau^2), \qquad a \sim N(0, \sigma_a^2), \qquad \theta \perp a. \tag{1}$$

Theorem 1 and Proposition 9 hold for general distributions.

A candidate's value to the institution is

$$v = \theta + \alpha(A)\, a, \qquad \alpha'(A) < 0, \tag{2}$$

where $A \in [0, 1]$ indexes AI capability. AI substitutes for execution, so the marginal value of execution skill declines with A. We use $\alpha(A) = \alpha_0(1 - A)$ in numerical illustrations; all

results require only monotonicity.

3.2 Hard and Soft Evidence

The institution learns about candidates through evidence, which can be of two kinds.

Hard evidence is publicly observed and cannot be manipulated by its provider. Execution skill generates hard evidence: work products reveal a perfectly and at negligible cost, for both groups. Conceptual capital generates hard evidence only through public output,

$$y = \theta + \eta, \qquad \eta \sim N\big(0, \rho_y^{-1}\big), \tag{3}$$

where ρ_y is the precision with which the institution's evaluators can assess a candidate's public work. Section 8 adds an AI-based evaluation T as a further source of hard evidence.

Soft evidence is observed privately by a referee and transmitted through unverifiable messages. Each candidate has one referee, who observes

$$s = \theta + \varepsilon, \qquad \varepsilon \sim N\big(0, \rho_s^{-1}\big), \tag{4}$$

independently of y. The referee of an insider is her *sponsor*, a member of the institution. The referee of an outsider is a faculty member at an outside institution. We deliberately give both referees *identical* information, so that the asymmetry we derive arises from incentives and not from access.[1]

3.3 The Institution

The institution has N members. It appoints the K candidates with the highest expected value given all credible information, subject to the procedural norm described in Section 4. Its members share a common reputation, which is increasing in the average quality of appointees; Section 9 models reputation explicitly. For now, it suffices that a member's payoff from an appointment of expected value w, relative to the marginal appointee at the cutoff c, is

$$u_j = \pi\, \mathbf{1}\{\text{own candidate}\} + \frac{\gamma}{N}\,(w - c), \tag{5}$$

where $\pi > 0$ is the private benefit of placing one's own candidate (collaboration, the continuation of a research program, intellectual legacy), γ is the value of reputation, and

[1] Allowing insiders' referees to be better informed, as is plausible, reinforces every result.

$1/N$ is the share of reputational gains and losses borne by each member.

An outside referee obtains a payoff $\pi_O > 0$ if her candidate is appointed and zero otherwise. She has no stake in the elite institution's reputation.

4 Who Can Be Believed? Verifiability and Communication

4.1 Outside Referees

The communication game between an outside referee and the institution has a simple structure. The referee observes s and sends a message $m \in M$, and the institution decides whether to appoint. Notice that the referee's payoff depends only on the decision and not on the state, so whatever she has observed, she prefers the message that maximizes the probability of appointment.

Lemma 1 (Outside soft information is irrelevant)**.** *In every perfect Bayesian equilibrium of the communication game between an outside referee and the institution, the probability that an outsider is appointed, conditional on her hard evidence, does not depend on the referee's soft signal s. The outcome is therefore the same as in an equilibrium in which the institution observes only hard evidence.*

Lemma 1 is a stark version of a familiar intuition: letters from outside are discounted. Its force here is that the discount is complete and arises from incentives alone. An outsider is evaluated only on hard evidence.

4.2 Sponsors

A sponsor's preferences, in contrast, depend on the state: by (5), she gains from appointing her candidate if and only if $\pi + (\gamma/N)(w - c) \geq 0$, where w is her posterior expectation of the candidate's value. Her interests are partly aligned with the institution's, and partial alignment supports informative communication.

We assume that the institution operates under a norm of *reciprocal deference*: each member's recommendation regarding her own candidate is followed. Such norms are sustained in repeated interaction among members who would each lose more from the breakdown of deference than they would gain from overriding a colleague (cf. Dessein, 2002, on delegation within organizations). Under reciprocal deference, the sponsor effectively decides.

Proposition 1 (Sponsorship). *Under reciprocal deference, a sponsor recommends her candidate if and only if*

$$W \;\geq\; c - b^*, \qquad b^* = \frac{N\pi}{\gamma}, \tag{6}$$

where $W = \mathbb{E}[v \mid a, y, s]$. *Consequently, the institution's information about insiders is the experiment* $\mathcal{E}_I = (a, y, s)$, *while its information about outsiders is* $\mathcal{E}_O = (a, y)$. *In particular,* $\mathcal{E}_O$ *is a garbling of* $\mathcal{E}_I$, *and the posterior precisions about* θ *are*

$$\rho_I = \rho_y + \rho_s, \qquad \rho_O = \rho_y. \tag{7}$$

Two features of this result deserve emphasis. First, the Blackwell ranking $\mathcal{E}_I \succeq \mathcal{E}_O$ is derived rather than assumed: both experiments are generated by the same physical evidence, but only one referee can credibly transmit what she knows. Second, the sponsor's information comes bundled with a bias b^*. Information and favoritism are joint products of sponsorship.

Remark 1 (Communication without deference). Without deference, a sponsor's recommendation is followed only if the institution's posterior, conditional on a recommendation, exceeds the cutoff. With a normal posterior increment of standard deviation Δ, this credibility constraint requires the standardized bias b^*/Δ to lie below a bound that is decreasing in the institution's selectivity. Highly selective institutions can therefore extract information from sponsors only if they defer, which provides a rationale for the norm. See Dessein (2002) for the general trade-off between delegation and communication.

To simplify expressions, we denote by $\beta_g \equiv \tau^2\rho_g/(1+\tau^2\rho_g)$ the weight on group g's evidence about θ, so that $\beta_I > \beta_O$ whenever $\rho_s > 0$. The posterior mean of value is

$$W_g = \alpha a + \mathbb{E}[\theta \mid \mathcal{E}_g] \;\sim\; N\big(0,\, V_g\big), \qquad V_g(\alpha) = \alpha^2\sigma_a^2 + \beta_g\tau^2. \tag{8}$$

We refer to V_g as the group's *visibility*. The institution appoints the K candidates with the highest score $W + b^*\mathbf{1}\{g = I\}$, so the cutoff c satisfies

$$m_I\left[1 - \Phi\left(\frac{c - b^*}{\sqrt{V_I}}\right)\right] + m_O\left[1 - \Phi\left(\frac{c}{\sqrt{V_O}}\right)\right] = K. \tag{9}$$

To focus on the interesting case where the institution is selective, we make the following assumption.

Assumption 1 (Selectivity). $K < (m_I + m_O)/2$.

Let P_g denote group g's probability of appointment, and $S_O \equiv m_O P_O / K$ the outsiders' share of appointments.

5 Informational Exclusion and the Favoritism Multiplier

5.1 A Blackwell Foundation

We first show that the insiders' informational advantage leads to their over-representation at selective thresholds under weak conditions. Let F_g denote the distribution of the posterior mean $W_g = \mathbb{E}[v \mid \mathcal{E}_g]$ for an arbitrary prior over ω and arbitrary experiments.

Definition 1 (Rotation)**.** F_I is a *rotation* of F_O at t^* if $F_I(t) \geq F_O(t)$ for $t < t^*$ and $F_I(t) \leq F_O(t)$ for $t > t^*$.

Theorem 1 (Blackwell exclusion)**.** *Suppose $\mathcal{E}_O$ is a garbling of $\mathcal{E}_I$ with respect to ω. Then for every $\alpha \geq 0$:*

(i) W_I dominates W_O in the convex order, and $W_O = \mathbb{E}[W_I \mid \mathcal{E}_O]$;

(ii) for every cutoff c, the institution's expected option value from screening an insider exceeds that from screening an outsider: $\mathbb{E}[(W_I - c)^+] \geq \mathbb{E}[(W_O - c)^+]$;

(iii) if F_I is a rotation of F_O at t^*, then $P_I \geq P_O$ for every common cutoff $c \geq t^*$; under Gaussian evidence $t^* = 0$, and the inequality is strict whenever $\rho_s > 0$ and $c > 0$;

(iv) if a has a continuous distribution and $\mathbb{E}[\theta^2] < \infty$, then as $\alpha \to \infty$ the difference between the groups' probabilities of appointment at the market-clearing cutoff vanishes.

Parts (i)–(iii) say that any Blackwell advantage, however it arises, shows up as an advantage in the upper tail. Part (iv) says that a large verifiable component, observed equally for all, makes the advantage immaterial. AI moves the economy from the regime of part (iv) toward that of part (iii).

The theorem has a decision-theoretic reading. The institution's problem is an option: it appoints only candidates whose posterior exceeds c, so its payoff is convex in the posterior mean. Blackwell's theorem states that more informative experiments are more valuable for every convex payoff, and Theorem 1 translates this into a statement about who is appointed. A group that is better understood is a better option, and selective institutions exercise options.

5.2 Comparative Statics under Gaussian Evidence

Two statistics summarize the allocation under Gaussian evidence: the visibility ratio and the standardized premium,

$$k(\alpha) \equiv \frac{V_I(\alpha)}{V_O(\alpha)} = \frac{\alpha^2\sigma_a^2 + \beta_I\tau^2}{\alpha^2\sigma_a^2 + \beta_O\tau^2} \geq 1, \qquad \tilde{b}(\alpha) \equiv \frac{b^*}{\sqrt{V_O(\alpha)}}. \tag{10}$$

Dividing all scores by $\sqrt{V_O}$ preserves rankings: outsiders' standardized scores are $N(0,1)$, and insiders' are $N(0,k)$ shifted by $\tilde{b}$. Thus $(k,\tilde{b})$ is sufficient for the allocation.

Lemma 2. *k is strictly decreasing in α, with $k(0) = \beta_I/\beta_O$ and $k \to 1$ as $\alpha \to \infty$. Hence k is strictly increasing in A.*

Proposition 2 (Informational exclusion)**.** *Let $b^* = 0$ and suppose Assumption 1 holds. Then $P_O < P_I$, and the outsider share S_O is strictly decreasing in A.*

The intuition behind this result turns on the upper tail. Bayesian shrinkage compresses noisier evidence toward the prior mean, so the distribution of outsiders' posteriors has less mass above a selective threshold. Execution skill, equally visible for both groups, dilutes this asymmetry; as its return vanishes, the asymmetry in the visibility of conceptual capital governs selection. The result requires neither prejudice nor any difference in ability. An institution staffed by unbiased Bayesians becomes more insular solely because the remaining object of selection is less legible from outside, and, by Lemma 1, it is illegible precisely because outside reports cannot be believed.

Proposition 3 (Favoritism multiplier)**.** *For any $b^* > 0$, the allocation under capability A coincides with that under visibility ratio $k(\alpha(A))$ and premium*

$$b^{\text{eq}}(A) = b^*\sqrt{\frac{V_O(\alpha(0))}{V_O(\alpha(A))}} > b^* \quad \text{for } A > 0, \tag{11}$$

evaluated at pre-AI visibility. Moreover, S_O is strictly decreasing in $\tilde{b}$. Hence AI amplifies the exclusionary effect of any fixed premium.

Now we give a numerical example. Table 1 reports the outsider share for $\tau = \sigma_a = \alpha_0 = 1$, $\rho_y^{-1/2} = 1.5$ and $(\rho_y + \rho_s)^{-1/2} = 0.5$, with outsiders making up 80% of candidates and an appointment rate of 5%. Without any premium, the outsider share falls by twenty percentage points as AI capability rises from zero to full automation. A premium of $b^* = 0.3$ costs outsiders ten points before AI and nineteen points under full automation, because its standardized size more than doubles, from 0.26 to 0.54.

Table 1: Outsider share of appointments, S_O: baseline model

AI capability A	$b^* = 0$	$b^* = 0.3$	Standardized premium $\tilde{b}$
0 (pre-AI)	70.7%	60.6%	0.26
0.5	60.8%	45.4%	0.40
0.8	52.7%	34.2%	0.51
1 (full automation)	50.3%	31.1%	0.54

Notes: Parameters are illustrative, not calibrated: $\tau = \sigma_a = \alpha_0 = 1$, outsider posterior noise 1.5, insider posterior noise 0.5, $m_O/(m_I + m_O) = 0.8$, appointment rate 5%. Entries are computed exactly from (9).

6 The Collapse of Apprenticeship: Formation and Revelation

So far, we have treated the pool of outsiders as fixed. Historically, however, the most important route by which outsiders became insiders was apprenticeship, that is, execution work for members of the institution. Apprenticeship performed two distinct functions. It *formed* conceptual capital, because judgment is acquired by working alongside those who exercise it. And it *revealed* conceptual capital, because an apprentice acquires a sponsor who observes her soft signal and can credibly transmit it. Now we show that AI, by dissolving the demand for execution labor, dismantles both functions at once.

6.1 Setup

Before selection, members hire a mass $D(A)$ of outsiders as apprentices, with $D'(A) < 0$ because AI substitutes for the execution tasks apprentices perform. Apprentices are selected on hard evidence of execution skill: an outsider is hired if $a \geq \bar{a}(A)$, where $m_O[1 - \Phi(\bar{a}/\sigma_a)] = D(A)$. Apprenticeship has two effects.

- *Formation:* the apprentice's conceptual capital rises from θ to $\theta + \lambda$, with $\lambda \geq 0$. Insiders, trained within the institution, receive the same increment.
- *Revelation:* the apprentice's employer becomes her sponsor, so her evidence becomes (a, y, s), and she receives the premium b^*.

Outsiders are thus divided into apprentices, of mass $D(A)$, who are informationally indistinguishable from insiders, and non-apprentices, of mass $m_O - D(A)$, who are evaluated on hard evidence alone. Let P_{ap} and P_{na} denote their probabilities of appointment.

Proposition 4 (Decomposition)**.** *The outsider share satisfies*

$$\frac{dS_O}{dA} = \underbrace{\frac{D'(A)\,(P_{ap} - P_{na})}{K}}_{\text{extensive margin}} + \underbrace{\frac{D(A)\,\partial_A P_{ap} + (m_O - D(A))\,\partial_A P_{na}}{K}}_{\text{intensive margin}}. \tag{12}$$

Whenever apprentices are more likely to be appointed than non-apprentices, the collapse of apprenticeship reduces the outsider share, independently of the visibility effect of Proposition 2. Moreover, the mean conceptual capital of insiders relative to outsiders is

$$\mu_I - \mu_O(A) = \lambda \left(1 - \frac{D(A)}{m_O}\right), \tag{13}$$

which is increasing in A.

The second part of this result is the deeper point. The collapse of apprenticeship creates a *genuine* difference in conceptual capital between the groups. An observer who compares insiders and outsiders after the fact will find insiders to be better on average, and will be correct. The institution's exclusion of outsiders thereby becomes statistically justified, even though it originates in the disappearance of a training channel rather than in any primitive difference between the groups.

6.2 Discouraged Investment

The formation channel interacts with candidates' own incentives to invest. Suppose that, before evaluation, each candidate chooses an investment $x_\theta \geq 0$ in conceptual capital and $x_a \geq 0$ in execution skill at convex cost $C(x_\theta + x_a)$, raising θ and a one-for-one. The institution anticipates equilibrium investment and adjusts its priors, as in signal-jamming models (Holmström, 1999). Let U denote the value of appointment and f_g the density of group g's score at its cutoff.

Proposition 5 (Discouraged investment)**.** *In equilibrium, the marginal return to investment in conceptual capital for a candidate in group g is $U\beta_g f_g$, and the marginal return to investment in execution skill is $U\alpha f_g$. Suppose $b^* = 0$ and $c^2 \geq V_I$. Then:*

(i) outsiders' marginal return to conceptual investment is strictly below insiders';

(ii) the ratio of outsiders' to insiders' largest marginal return to investment is f_O/f_I when $\alpha \geq \beta_I$, and $(\beta_O f_O)/(\beta_I f_I)$ when $\alpha \leq \beta_O$; the latter is strictly smaller.

The intuition behind this result is as follows. Before AI, an outsider could invest in a skill whose returns were fully visible regardless of her origin. After AI, the only rewarding skill is one whose returns are attenuated by the shrinkage factor β_O. The outsider's incentive to invest falls relative to the insider's, outsiders invest less, and the institution's expectation that outsiders are weaker is confirmed. This is the logic of Coate and Loury (1993), here triggered not by beliefs but by a change in the verifiability of the skill that matters.

7 The Cheapening of Patronage: Influence Activities

Sponsorship creates a market. If a sponsor's recommendation determines appointment, candidates have an incentive to influence it, and the means of influence need not be intellectual. A candidate may supply services of personal value to her sponsor: administrative work, accommodation of the sponsor's schedule, or help with matters unrelated to research. Following Milgrom (1988), we call these *influence activities*. They require proximity, so they are available to insiders and apprentices but not to other outsiders.

7.1 Setup

Each proximate candidate has an influence efficiency $e \geq 0$, distributed according to G_e independently of (θ, a). Providing services of value t to her sponsor costs her t/e. The sponsor's private benefit from appointing the candidate becomes $\pi + t$. Let U denote the candidate's value of appointment.

Proposition 6 (Influence as a random premium)**.** *A proximate candidate with posterior value W is appointed if and only if*

$$W + b^* + B \;\geq\; c, \qquad B \equiv \frac{NU}{\gamma}\, e. \tag{14}$$

Equivalently, a candidate with $W < c - b^*$ is appointed if and only if her efficiency exceeds

$$e^*(W) = \frac{\gamma\,(c - b^* - W)}{NU}. \tag{15}$$

In standardized units $z = W/\sqrt{V_I}$ and $\zeta = (c - b^*)/\sqrt{V_I}$,

$$e^*(z) = \frac{\gamma}{NU}\sqrt{V_I(\alpha(A))}\,(\zeta - z), \tag{16}$$

so that, holding the standardized cutoff fixed, the efficiency required to bridge any given gap in rank falls with A by the factor $\sqrt{V_I(\alpha(A))/V_I(\alpha(0))}$. The value of services required, $t^* = \gamma(c - b^* - W)/N$, falls by the same factor.

Influence thus acts as a random, privately purchased premium that is unrelated to merit. Its effect is subject to exactly the multiplier of Proposition 3: the compression of the selection index makes favor cheaper. In our numerical example, full automation lowers the required efficiency at every standardized rank by one-third.

7.2 The Composition of Entrants

We distinguish three kinds of appointee: *conceptual talents*, whose posterior on θ alone clears the cutoff; *execution talents*, who clear it by virtue of αa; and *cultivators*, who clear it only through B.

Proposition 7 (Composition)**.** *Holding the standardized cutoff fixed:*

(i) as $A \to 1$, the mass of execution talents vanishes;

(ii) at every standardized rank below the cutoff, the minimum efficiency of a cultivator falls with A, so the set of cultivators expands from those with exceptionally high e to all those with e above a strictly lower threshold;

(iii) the expected conceptual capital of cultivators is below that of other appointees, so their rising share lowers the average quality of appointees relative to the symmetric-information benchmark.

Proposition 7 gives a precise prediction about who enters. Before AI, the institution admits conceptual talents, execution talents and a small number of exceptionally skilled cultivators. After AI, it admits conceptual talents and a broader class of moderately skilled cultivators. The execution talents, historically the route of the able outsider, are the ones displaced.

Table 2 reports simulations that combine all channels. With every channel active, the outsider share falls from 70% to 32%, and the quality shortfall relative to the symmetric-information benchmark rises from 0.26 to 0.40 standard deviations of conceptual capital. Notice that, before AI, apprenticeship slightly *raises* the outsider share, because apprentices acquire sponsors; under full automation, it lowers the share by four points, because the training gap is no longer offset by the revelation of apprentices. The share of appointees who enter only through influence rises from 20.5% to 22.0%.

Table 2: Outsider share and quality with all channels

	Outsider share S_O				
A	Baseline	+ Influence	+ Apprenticeship	All channels	Quality shortfall
0	70.7%	63.4%	72.1%	70.1%	0.26
0.5	60.8%	49.6%	60.0%	52.5%	0.37
0.8	52.7%	39.0%	50.4%	38.1%	0.39
1	50.3%	36.1%	46.3%	32.4%	0.40

Notes: Parameters as in Table 1, with $b^* = 0$ to isolate the new channels. The baseline column is exact; the other columns are Monte Carlo averages over 3×10^6 candidates (sampling error about 0.3 points). *Influence: B* is exponentially distributed with mean 0.2 among proximate candidates. *Apprenticeship:* the apprenticeship mass is $D(A) = 0.08(1 - A)$ of the population; apprentices are the outsiders with the highest a; insiders and apprentices receive the formation increment $\lambda = 0.3$ (known to the institution) and are evaluated on (a, y, s). *Quality shortfall:* the difference in mean θ of appointees, in units of τ, between a benchmark in which every candidate is evaluated on (a, y, s) without influence, holding formation fixed, and the all-channels economy.

8 AI as a Verifier

So far, we have treated AI purely as a substitute for execution. AI is also a potential technology of *evaluation*. In this section, we ask under what conditions it reverses the forces identified above.

8.1 A Common Public Test

Suppose AI provides a publicly observed evaluation of conceptual capital, $T = \theta + \xi$ with $\xi \sim N(0, \rho_T(A)^{-1})$ and $\rho_T'(A) \geq 0$. Being hard evidence, T is available for both groups. Public output is subject to two opposing forces: AI enables outsiders to produce and disseminate work, but it also multiplies the volume of submissions competing for the evaluators' finite attention H. We write

$$\rho_y(A)^{-1} = \sigma_0^2 - d(A) + g\big(n(A)/H\big), \tag{17}$$

with $d' > 0$ (production), and $n' > 0$ and $g' > 0$ (congestion). Insiders' soft channel is insulated from congestion. The precisions become $\rho_O = \rho_y + \rho_T$ and $\rho_I = \rho_y + \rho_T + \rho_s$.

Proposition 8 (Opening versus closure)**.** *Let $b^* = 0$.*

(i) The insiders' experiment remains Blackwell-superior for every ρ_T, yet $k \to 1$ and $S_O \to m_O/(m_I + m_O)$ as $\rho_T \to \infty$.

(ii) Writing $x = \alpha^2\sigma_a^2$, the institution becomes more closed at the margin, $dS_O/dA < 0$, if and only if

$$\frac{d}{dA}\log\left(\beta_I - \beta_O\right) \;>\; \frac{d}{dA}\log\left(x + \beta_O\tau^2\right), \tag{18}$$

where $\beta_I - \beta_O = \tau^4\rho_s/\left[(1+\tau^2\rho_I)(1+\tau^2\rho_O)\right]$.

Part (i) separates two notions that are often confused. A common verifiable test does not eliminate the insiders' informational advantage in the Blackwell sense, but it makes that advantage immaterial for appointments. Part (ii) states the race between the two effects of AI. AI closes institutions when it erodes the verifiable component of value faster than it builds verifiable channels for ideas, or when the congestion of public channels outpaces both.

8.2 Designable Tests and Persuasion

A public test may also be designable. The outsider's home institution may choose the protocol by which its students' ideas are assessed, such as which projects to submit, which benchmarks to apply and how results are reported, and commit to it, for instance by delegating evaluation to an AI system whose specification is public. The home institution maximizes placement, and the elite institution is Bayesian. This is a problem of Bayesian persuasion (Kamenica and Gentzkow, 2011).

To isolate the persuasion logic, we take $A = 1$, so $v = \theta$, let the outsider's conceptual capital have an arbitrary distribution G with continuous density and mean μ_O, and suppose the elite institution appoints an outsider if and only if her posterior mean is at least $c > \mu_O$, taking c as given.

Proposition 9 (Verifiability beats precision)**.** *The placement-maximizing verifiable test is a pass–fail standard that passes the candidate if and only if $\theta \geq \hat{\theta}$, where*

$$\mathbb{E}[\theta \mid \theta \geq \hat{\theta}] = c. \tag{19}$$

It places a fraction $q^* = 1 - G(\hat{\theta})$ of outsiders. Moreover,

$$q^* \;>\; 1 - G(c) \;>\; 0, \tag{20}$$

where $1 - G(c)$ is placement under full disclosure of θ and 0 is placement under any soft channel (Lemma 1).

Table 3: Outsider placement under alternative evaluation regimes

Cutoff c (s.d. above mean)	Soft channel	Full disclosure	Optimal verifiable test
0.5	0	30.9%	69.8%
1.0	0	15.9%	38.1%
1.5	0	6.7%	16.6%
2.0	0	2.3%	5.8%

Notes: Standard normal prior over outsiders' conceptual capital; the elite institution appoints if the posterior mean is at least c, taken as fixed.

The comparison is stark. A perfectly precise but soft assessment of an outsider's ideas is worthless to her, while a coarse but verifiable one is worth more than perfect precision would be. Table 3 reports placement rates for a standard normal prior: for a cutoff one standard deviation above the outsider mean, the optimal test more than doubles placement relative to full disclosure, from 15.9% to 38.1%.

Remark 2 (Equilibrium and competition). With a fixed capacity, persuasion by outsiders' institutions raises the elite institution's cutoff, which partly offsets the gains in placement; the elite institution is Bayesian, so its expected surplus from a persuaded marginal appointee is zero. When many outside institutions design tests competitively, competition in persuasion pushes toward more disclosure (Gentzkow and Kamenica, 2017), and the problem acquires the structure of the grading and certification models of Ostrovsky and Schwarz (2010) and Boleslavsky and Cotton (2015).

Remark 3 (Manipulation). An AI evaluator that candidates can game generates evidence whose informativeness depends on how much candidates differ in their ability to game it (Frankel and Kartik, 2019). If the ability to game is correlated with proximity, as influence efficiency is, manipulation reintroduces the insiders' advantage through the verifiable channel. The design of evaluation that is robust to manipulation is therefore part of the problem of mobility.

Remark 4 (AI as a generator of ideas). If AI also generates ideas, we can decompose conceptual capital as $\theta = \iota + j$, where ι is idea generation and j is judgment about which ideas to pursue. Substitution of ι lowers its weight in value exactly as substitution of a does. If, as seems plausible, judgment is less verifiable than generation, the analysis applies a fortiori: the remaining object of selection becomes still less legible.

9 Reputation

9.1 The Reputational Commons

The premium $b^* = N\pi/\gamma$ of Proposition 1 reflects an externality: each member bears only a share $1/N$ of the reputational cost of a weak appointment.

Proposition 10 (Reputational commons)**.** *An institution that fully internalized reputational effects would set the premium $b^{FB} = \pi/\gamma$, and $b^{FB} = 0$ if private benefits carry no weight for the institution. The equilibrium premium exceeds b^{FB} by the factor N. With influence activities, the effective premium $b^* + B$ is increasing in N in the sense of first-order stochastic dominance.*

Larger institutions are therefore more susceptible to favoritism and to influence, not because their members are more partial, but because each member internalizes less of the reputational consequence.

9.2 Dynamics

External audiences, such as prospective students, funders, journals and employers, update the institution's reputation from the realized quality of appointees. Let Q_t be the expected conceptual capital of the cohort appointed at t, and

$$R_{t+1} = (1-\delta)R_t + \delta Q_t. \tag{21}$$

Proposition 11 (Delayed decline)**.** *Let $Q^{\text{sym}}(A)$ denote appointee quality under symmetric information and no premium, and $L(A) \equiv Q^{\text{sym}}(A) - Q(A) \geq 0$ the selection loss. Following a permanent increase in A, reputation converges geometrically, at rate $1-\delta$, to $R^* = Q^{\text{sym}}(A) - L(A)$. The selection loss L is increasing in the standardized premium and in the mass of influence entrants.*

Two features of the transition matter. First, there is a *reputational overhang*: for a period of order $1/\delta$, incumbents enjoy a reputation accumulated under a more open regime, which weakens the pressure for reform. Second, the formation channel partly *masks* the decline. Because insiders really do acquire more conceptual capital (Proposition 4), the institution's appointees remain strong in absolute terms even as the selection loss L grows. Audiences that compare the institution with outsiders, rather than with what the institution could achieve, will detect the decline late.

Proposition 12 (Entry)**.** *Suppose a challenger commits to evaluating outsiders through verifiable channels, raising its precision about outsiders to $\rho_O^C > \rho_O$, and admits no influence. If its steady-state quality exceeds the incumbent's, $Q^C > Q^E$, its reputation eventually exceeds the incumbent's, regardless of initial reputations.*

The theory therefore predicts not permanent entrenchment but a slow rotation of elite status toward institutions that keep legible channels of entry. The welfare cost is concentrated in the transition: a generation of misallocated talent, and the ideas it would have pursued.

10 Empirical Implications

Our model yields predictions that distinguish it from accounts based only on rising competition.

P1. *Declining outsider share.* The share of entrants to elite doctoral programs and research positions who come from non-elite institutions falls after the diffusion of generative AI, more strongly in fields where AI substitutes more heavily for execution.

P2. *Collapse of apprenticeship.* Entry-level execution positions for outsiders contract before changes in appointment outcomes become visible, and the outsider share among appointees falls most where apprenticeship was the dominant route of entry.

P3. *Heterogeneity by evaluation format.* The decline is smaller where selection relies on verifiable work products than where it relies on letters and interviews (Lemma 1, Proposition 9).

P4. *Size and insularity.* Insider shares and the dispersion of appointee quality rise more in larger institutions (Proposition 10).

P5. *Outcome test.* Under pure informational exclusion, the expected performance of *marginal* insider and outsider appointees is equal, even when formation makes *average* insiders better. Under favoritism or influence, marginal outsiders outperform. This Becker-type test distinguishes between mechanisms and is robust to the formation channel.

P6. *Composition.* After AI, the lower tail of insider appointees thickens, and lower-tail quality is correlated with measures of service-oriented proximity to sponsors (Proposition 7).

11 Conclusion

In this paper, we studied how generative AI changes selection into elite institutions. We showed that the economics of access to such institutions is, in large part, the economics of verifiability. AI devalues the most verifiable form of human capital and leaves selection resting on soft information that only insiders' sponsors can credibly transmit. Four consequences follow: outsiders are under-selected even by unbiased evaluators, any given favoritism matters more, the dissolution of apprenticeship creates genuine and self-confirming group differences, and patronage becomes cheaper to obtain. Because reputation is shared among the members of an institution, these forces erode quality slowly, and the erosion is partly masked.

Our results also suggest where remedies lie. The decisive instrument is not more precise evaluation but more *verifiable* evaluation: Proposition 9 shows that a coarse verifiable standard can outperform perfect precision delivered through soft channels. Institutions can develop verifiable channels for conceptual capital, such as structured proposals assessed against explicit criteria or audited AI-based evaluation; rebuild apprenticeship around conceptual rather than executional tasks; limit discretion through blind initial review; protect public channels from congestion; disclose appointee outcomes by route of entry, so that members internalize reputational costs; and randomize among candidates near the threshold, where the expected cost in quality is small. Whether AI ultimately opens or closes elite institutions depends less on the technology itself than on whether it is used to verify ideas or merely to execute them.

A natural way to extend our model is to make the cutoff of the elite institution respond to persuasion by many competing outside institutions. Another extension is to model the dynamics of the candidate pool, in which today's appointees become tomorrow's sponsors, so that the composition of entrants feeds back into the information structure. Finally, another interesting direction is to test the outcome test of Section 10 using data on the careers of appointees by route of entry.

A Proofs

Proof of Lemma 1. Fix hard evidence h and an equilibrium. Let $p(m)$ be the probability of appointment following message m. The referee's payoff from m is $\pi_O p(m)$, which does not depend on s. Hence every message sent with positive probability by any type must attain $\bar{p} = \sup_m p(m)$, and appointment occurs with probability $\bar{p}$ regardless of s. The outcome coincides with that of the equilibrium in which the institution ignores messages and appoints with probability $\bar{p}$ given h. □

Proof of Proposition 1. Under reciprocal deference, the sponsor's recommendation determines appointment, so she recommends if and only if $\pi + (\gamma/N)(W - c) \geq 0$, that is, $W \geq c - N\pi/\gamma$. Her recommendation is a function of $W = \mathbb{E}[v \mid a, y, s]$, so appointments of insiders are determined by (a, y, s). For outsiders, Lemma 1 implies that appointments are determined by (a, y). Since (a, y) is a coordinate projection of (a, y, s), $\mathcal{E}_O$ is a garbling of $\mathcal{E}_I$. Gaussian updating with independent signals gives $\rho_I = \rho_y + \rho_s$ and $\rho_O = \rho_y$. □

Proof of Theorem 1. (i) Garbling means that $\mathcal{E}_O$ is conditionally independent of ω given $\mathcal{E}_I$. Hence $\mathbb{E}[v \mid \mathcal{E}_I, \mathcal{E}_O] = W_I$, and by the law of iterated expectations $W_O = \mathbb{E}[W_I \mid \mathcal{E}_O]$. Conditional Jensen's inequality gives $\mathbb{E}\phi(W_I) \geq \mathbb{E}\phi(W_O)$ for every convex ϕ. (ii) Apply (i) with $\phi(x) = (x - c)^+$. (iii) For $c \geq t^*$, $P_I = 1 - F_I(c) \geq 1 - F_O(c) = P_O$ by the definition of a rotation. Under Gaussian evidence, $F_g = N(0, V_g)$ with $V_I > V_O$ when $\rho_s > 0$, so the distributions cross exactly once, at 0, and the inequality is strict for $c > 0$. (iv) Write $W_g/\alpha = a + M_g/\alpha$ with $M_g = \mathbb{E}[\theta \mid \mathcal{E}_g]$. Since $\mathbb{E}[M_g^2] \leq \mathbb{E}[\theta^2] < \infty$, $M_g/\alpha \to 0$ in L^2, so W_g/α converges in distribution to a for both groups. The market-clearing cutoff, rescaled by α, converges to the corresponding quantile of a, and the continuity of the distribution of a implies that both groups' probabilities of appointment converge to the same limit. □

Proof of Lemma 2. Let $x = \alpha^2\sigma_a^2$, $p = \beta_I\tau^2$ and $q = \beta_O\tau^2$, with $p > q$. Then $k = (x + p)/(x + q)$ and $dk/dx = (q - p)/(x + q)^2 < 0$. Since x is increasing in α and $\alpha'(A) < 0$, k is increasing in A. □

Proof of Proposition 2. At $c = 0$ the institution would appoint $(m_I + m_O)/2 > K$, so $c > 0$. As $V_I > V_O$, $P_I > P_O$. Let $z = c/\sqrt{V_I} > 0$, so that $P_I = 1 - \Phi(z)$ and

$P_O = 1 - \Phi(\sqrt{k}\, z)$. Differentiating (9),

$$\frac{dz}{dk} = -\frac{m_O\, \phi(\sqrt{k}\, z)\, z/(2\sqrt{k})}{m_I \phi(z) + m_O \sqrt{k}\, \phi(\sqrt{k}\, z)} < 0.$$

Hence P_I increases in k. Since $m_I P_I + m_O P_O = K$, $m_O P_O$ and S_O decrease in k, and thus in A by Lemma 2. □

Proof of Proposition 3. Dividing scores by $\sqrt{V_O}$ preserves rankings; standardized outsider scores are $N(0,1)$ and insider scores are $N(0,k) + \tilde{b}$, so the allocation depends only on $(k, \tilde{b})$. Fixing the outsider cutoff, an increase in $\tilde{b}$ raises insider appointments above K, so the cutoff must rise, and P_O falls. Since V_O is increasing in α, $\tilde{b}$ increases in A, and equating standardized premia gives b^{eq}. □

Proof of Proposition 4. The identity $KS_O = DP_{ap} + (m_O - D)P_{na}$ follows from the partition of outsiders. Differentiating gives the decomposition, whose first term is negative when $D' < 0$ and $P_{ap} > P_{na}$. Mean conceptual capital is $\lambda D(A)/m_O$ among outsiders and λ among insiders, which gives the gap; it is increasing in A since $D' < 0$. □

Proof of Proposition 5. A deviation to investment x_θ shifts the candidate's posterior mean by $\beta_g(x_\theta - \hat{x}_\theta)$, where $\hat{x}_\theta$ is the anticipated investment, and leaves its distribution otherwise unchanged. A deviation in x_a shifts it by $\alpha(x_a - \hat{x}_a)$, since a is hard. The marginal effects on the probability of appointment are therefore $\beta_g f_g$ and αf_g. (i) With $b^* = 0$, $f_g(c) = \phi(c/\sqrt{V_g})/\sqrt{V_g}$, and $\partial \log f_g / \partial V_g = (c^2 - V_g)/(2V_g^2)$, which is nonnegative for $V_g \le V_I \le c^2$. Hence $f_I \ge f_O$ and, with $\beta_I > \beta_O$, $\beta_I f_I > \beta_O f_O$. (ii) Each candidate's largest marginal return is $U f_g \max\{\alpha, \beta_g\}$. If $\alpha \ge \beta_I$, the ratio is f_O/f_I; if $\alpha \le \beta_O$, it is $\beta_O f_O/(\beta_I f_I)$, which is strictly smaller since $\beta_O < \beta_I$. □

Proof of Proposition 6. By Proposition 1 with private benefit $\pi + t$, the sponsor recommends if and only if $W \ge c - N(\pi + t)/\gamma$. The candidate's smallest transfer that secures appointment is $t^* = \max\{0, \gamma(c - W)/N - \pi\}$, which is worthwhile if and only if $t^*/e \le U$. Hence she is appointed if and only if $W + b^* + NUe/\gamma \ge c$. Solving for e gives $e^*(W)$, and substituting $W = z\sqrt{V_I}$ and $c - b^* = \zeta\sqrt{V_I}$ gives the standardized expression, which scales with $\sqrt{V_I}$ and is therefore decreasing in A. □

Proof of Proposition 7. (i) Execution talents satisfy $\alpha a \geq c - \mathbb{E}[\theta \mid \mathcal{E}_g]$ while $\mathbb{E}[\theta \mid \mathcal{E}_g] < c$; as $\alpha \to 0$ with the standardized cutoff fixed, their mass converges to zero. (ii) This follows from Proposition 6. (iii) Cultivators are appointed with $W < c - b^*$, while other appointees have $W \geq c - b^*$. Since B is independent of ω,

$$\mathbb{E}[\theta \mid \text{cultivator}] = \mathbb{E}\left[\mathbb{E}[\theta \mid W] \,\middle|\, W < c - b^*,\, W + B \geq c - b^*\right] < \mathbb{E}\left[\mathbb{E}[\theta \mid W] \,\middle|\, W \geq c - b^*\right].$$

The displacement of higher-W candidates by cultivators lowers the mean quality of appointees relative to the benchmark. □

Proof of Proposition 8. (i) (a, y, T) is a projection of (a, y, T, s), so the Blackwell ranking persists. As $\rho_T \to \infty$, $\beta_I, \beta_O \to 1$ and $k \to 1$, and by Theorem 1(iii) and Proposition 2 the probabilities of appointment equalize. (ii) Since $k - 1 = (\beta_I - \beta_O)\tau^2/(x + \beta_O\tau^2)$, k increases in A if and only if the stated condition holds, and by Proposition 2, S_O moves inversely with k. The expression for $\beta_I - \beta_O$ follows from $\beta_g = \tau^2\rho_g/(1 + \tau^2\rho_g)$. □

Proof of Proposition 9. Fix any verifiable test, and let $\mathcal{A}$ be the event that the posterior mean is at least c, with $q = \mathbb{P}(\mathcal{A})$. Then $\mathbb{E}[\theta \mathbf{1}_{\mathcal{A}}] = \mathbb{E}[\mathbb{E}[\theta \mid \text{test}]\mathbf{1}_{\mathcal{A}}] \geq cq$. Among all events of probability q, the upper tail $\{\theta \geq G^{-1}(1-q)\}$ maximizes $\mathbb{E}[\theta \mathbf{1}_{\cdot}]$, so $\mathbb{E}[\theta \mid \theta \geq G^{-1}(1-q)] \geq c$. The left side is continuous and strictly decreasing in q, and it equals $\mu_O < c$ at $q = 1$. Hence $q \leq q^*$, where q^* solves the equation with equality. The pass–fail test with threshold $\hat{\theta} = G^{-1}(1-q^*)$ attains q^*. Full disclosure places $1 - G(c)$; since $\mathbb{E}[\theta \mid \theta \geq c] > c$, the conditional-mean equation implies $\hat{\theta} < c$, so $q^* > 1 - G(c)$. Placement under soft channels is zero because, by Lemma 1, the elite institution's posterior equals the prior mean $\mu_O < c$. □

Proof of Proposition 10. An institution that internalizes reputation replaces γ/N by γ in (5), which gives $b^{FB} = \pi/\gamma$. Since $B = NUe/\gamma$ is increasing in N for each e, $b^* + B$ increases in N in the sense of first-order stochastic dominance. □

Proof of Proposition 11. For $\delta \in (0, 1)$, the law of motion is a stable linear difference equation with fixed point $R^* = Q = Q^{\text{sym}} - L$, and deviations decay geometrically at rate $1 - \delta$. L increases in $\tilde{b}$ by Proposition 3 and in the mass of influence entrants by Proposition 7(iii). □

Proof of Proposition 12. Each institution's reputation converges to its own steady state, independently of initial conditions, so the ranking of the limits is the ranking of steady-state qualities. □

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